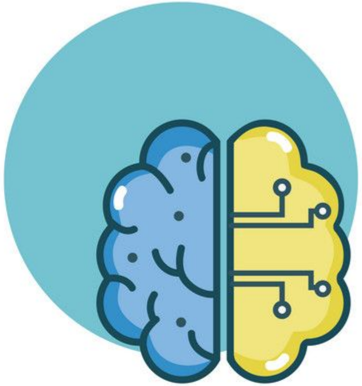
INTRODUCTION TO MACHINE LEARNING

Machine Learning Basics: Data, Features, Learning Types

Elisa Ricci

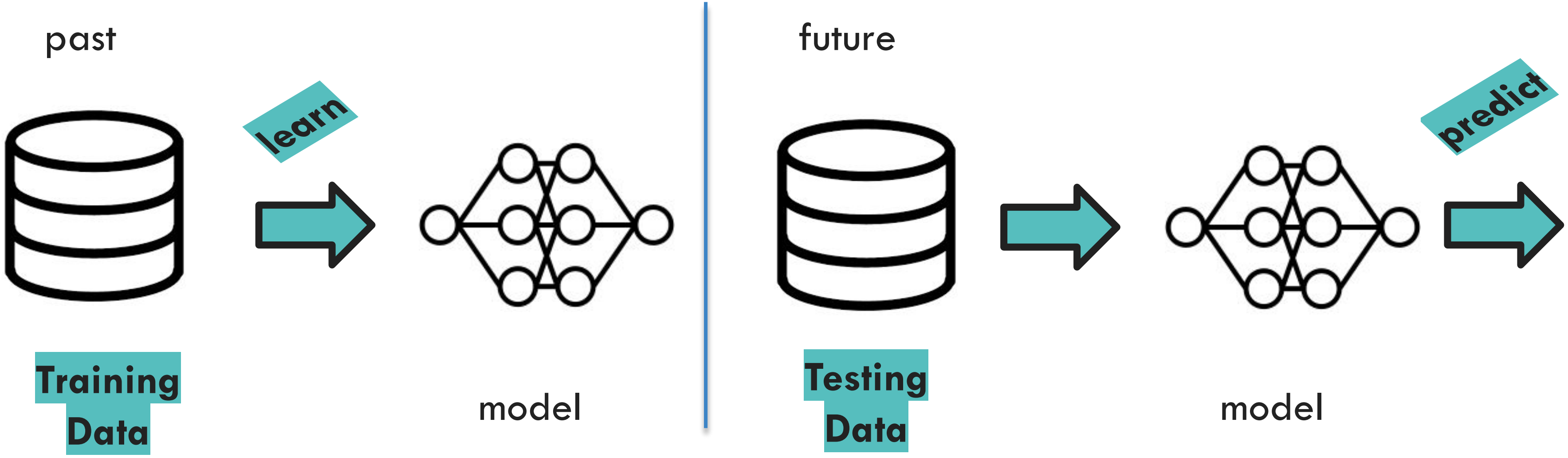


Machine Learning is…

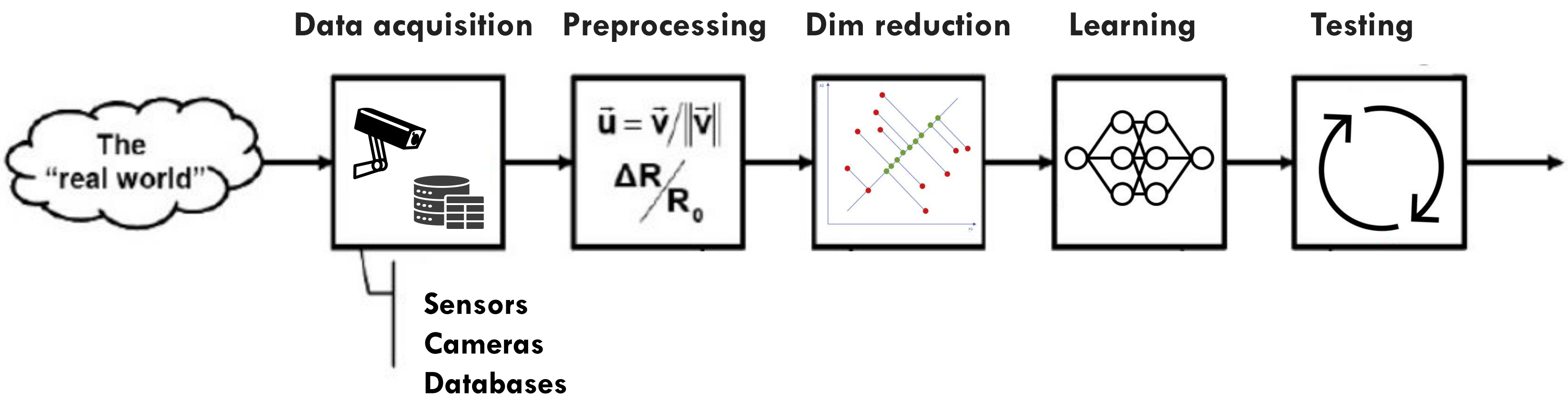
The goal of machine learning is to develop methods that can **automatically detect patterns in data**, and then to use the uncovered patterns to **predict future data** or other outcomes of interest.

Kevin P. Murphy

The Learning Process



The Learning Process



The Learning Process

Data Acquisition:

- This phase involves **collecting relevant data** for the problem at hand.
- This data can come from **various sources** such as databases or sensors.
- Ensuring the **quality and relevance** of the data is crucial at this stage.

The Learning Process

Data Preprocessing:

- Once the data is acquired, it often needs to be cleaned and prepared for analysis.
- This step involves **handling missing values, dealing with outliers,** and **transforming the data into a suitable format** for further analysis.
- Common preprocessing techniques include normalization, feature scaling, and handling categorical variables.

a. Employees

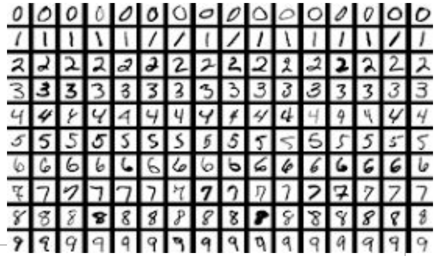
name	*age*	*salary*
Paul	1978	NULL
Paul	NULL	29,000
Paul	1979	NULL
Melanie	1990	NULL
Bob	NULL	37,000
Bob	1977	NULL
Charlie	1978	32,000

b. Employees

name	*age*	*salary*
Paul	?	29,000
Melanie	1990	NULL
Bob	1977	37,000
Charlie	1978	32,000

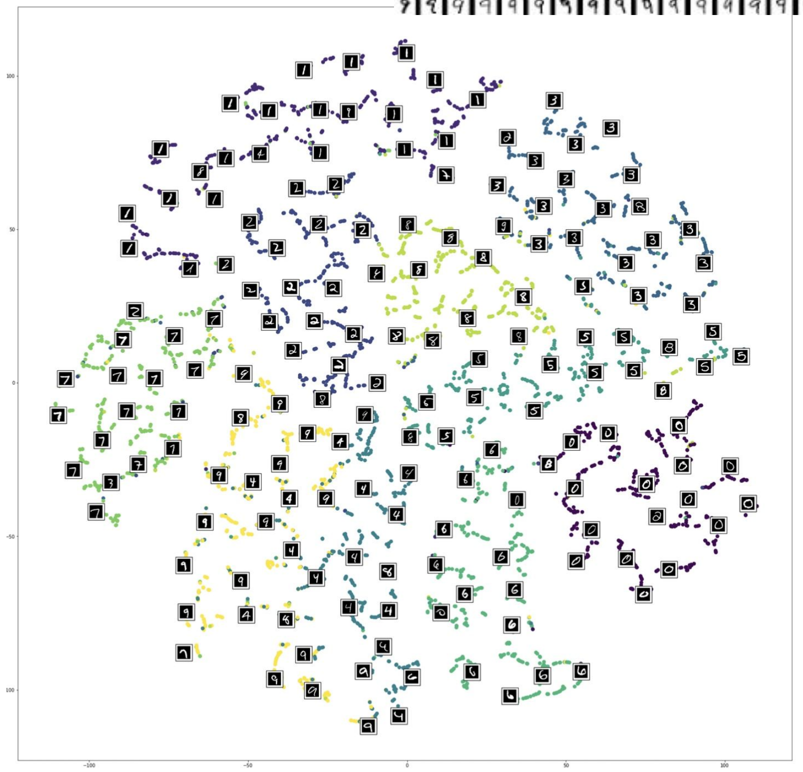
Paul.age = 1978 OR 1979 ?

The Learning Process



Dimensionality Reduction:

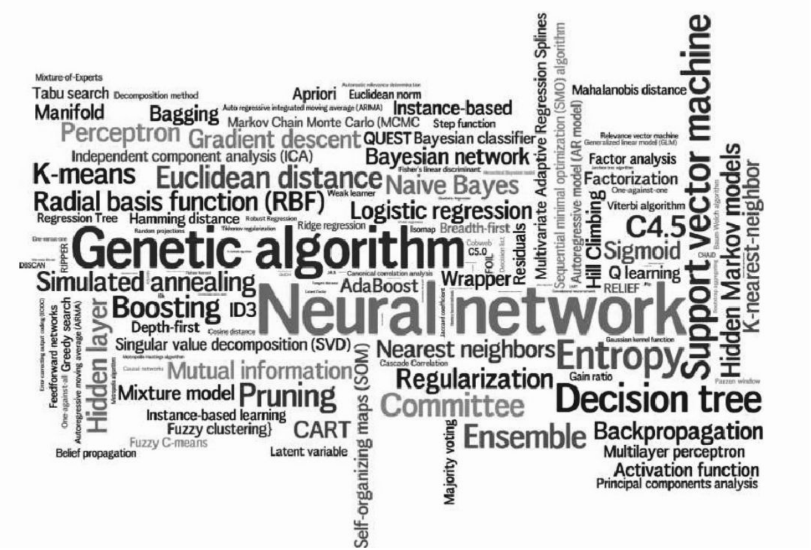
- Datasets may contain a large number of features (will discuss this later), which can lead to noise and increased computational complexity.
- Dimensionality reduction techniques can be applied to **reduce the number of features** while preserving the most important information.



The Learning Process

Model Learning:

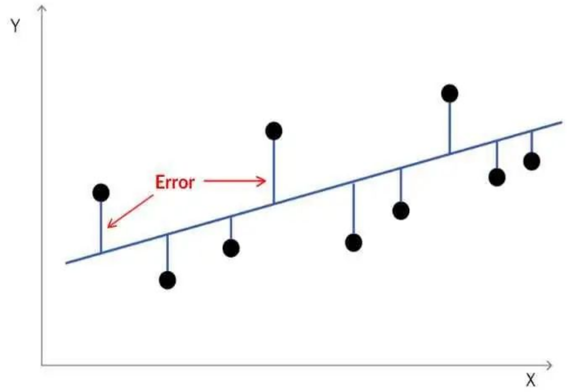
- This is the core of the ML pipeline, where **a model is trained on the preprocessed data** to learn patterns and relationships.
- The **choice of model** depends on the nature of the problem (e.g., classification, regression) and the characteristics of the data.
- Common machine learning **algorithms** include linear regression, decision trees, support vector machines, and neural networks.



The Learning Process

Model Testing:

- Once the model is trained, it needs to be evaluated to assess its performance and **generalization capabilities.**
- This is typically done using a separate datase called the **test set**, which was not used during the training phase.
- **Performance metrics** such as accuracy, precision, recall, F1-score, or mean squared error are computed to quantify the model performance.



The Learning Process

Throughout the pipeline, it is important to **iterate and refine** each step based on the insights gained from the evaluation phase. This iterative process helps improve the performance of the model and ensure its effectiveness in real-world applications.

Testing

Evaluation

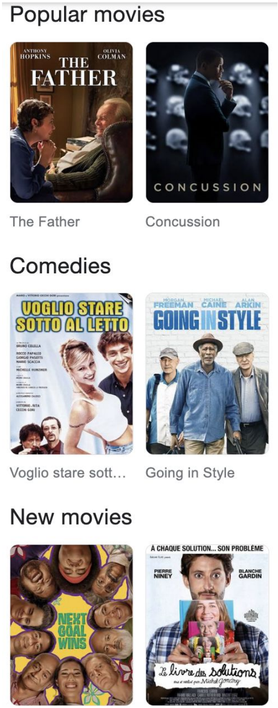
Model creation

Let's start with Data

Data

examples

Data



Data

Data

examples

Data

Data

examples

Data

examples

Data

Representing examples

What is an example? How is it **represented**?

Features

examples

features

$x^1_1, x^1_2, x^1_3, \dots, x^1_d$

$x^2_1, x^2_2, x^2_3, \dots, x^2_d$

$x^3_1, x^3_2, x^3_3, \dots, x^3_d$

Features

examples

features

grey, small, ...

brown, big, ...

black, small, ...

Features

- How our algorithms actually "view" the data
- **Features are** the **questions we can ask about the examples**
- Features in generally are **represented with vectors**

$$\mathbf{x} = [x^1_{1}, x^1_{2}, x^1_{3}, \ldots, x^1_{d}]$$

Features

- How our algorithms actually "view" the data
- **Features are** the **questions we can ask about the examples**
- Features in generally are **represented with vectors**

$$\mathbf{x} = [x^1_1, x^1_2, x^1_3, \ldots, x^1_d]$$

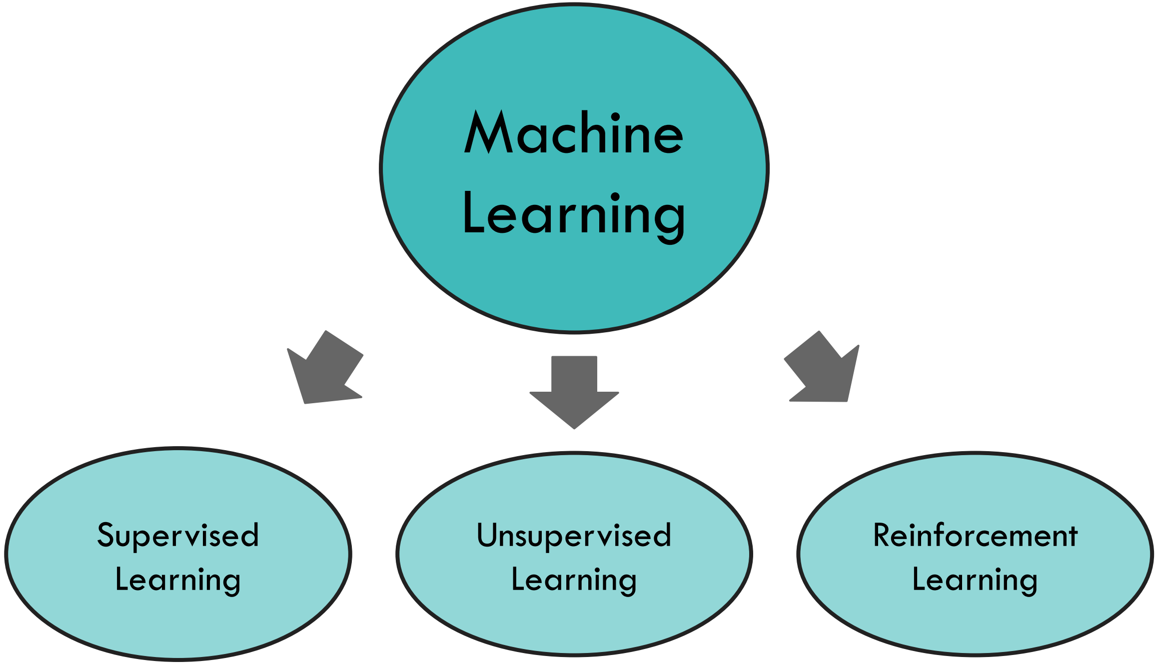
Any problem?

Take Home Message

Mapping data to features often implies a **loss of information** because features are typically derived or selected representations of the original data that capture certain aspects deemed relevant for a particular task or analysis.

This process involves **simplification or abstraction**, which can lead to a reduction in the amount of information available compared to the raw data.

Types of Learning



Supervised Learning

examples | **labels**

cat

dog

cat

Given **labeled examples…**

Supervised learning

examples | **labels**

$x^1_1, x^1_2, x^1_3, \ldots, x^1_d$ — y_1

$x^2_1, x^2_2, x^2_3, \ldots, x^2_d$ — y_2

$x^3_1, x^3_2, x^3_3, \ldots, x^3_d$ — y_3

Given **labeled examples…**

Supervised learning

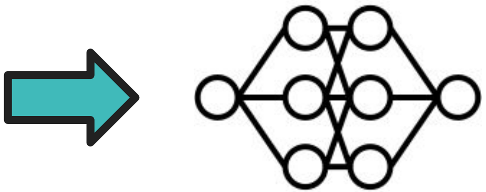
examples | **labels**

cat

dog

cat

model

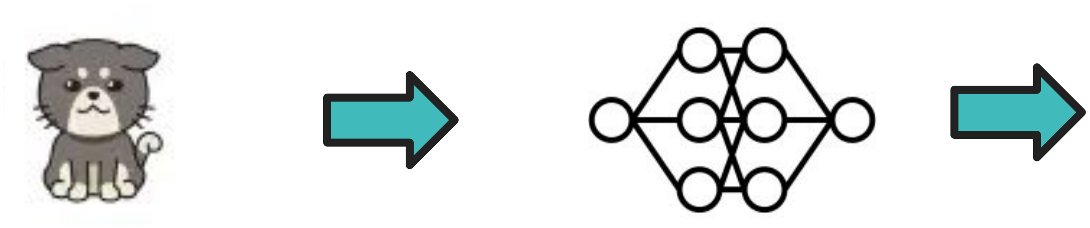


… build a **model** and…

Supervised learning

test example

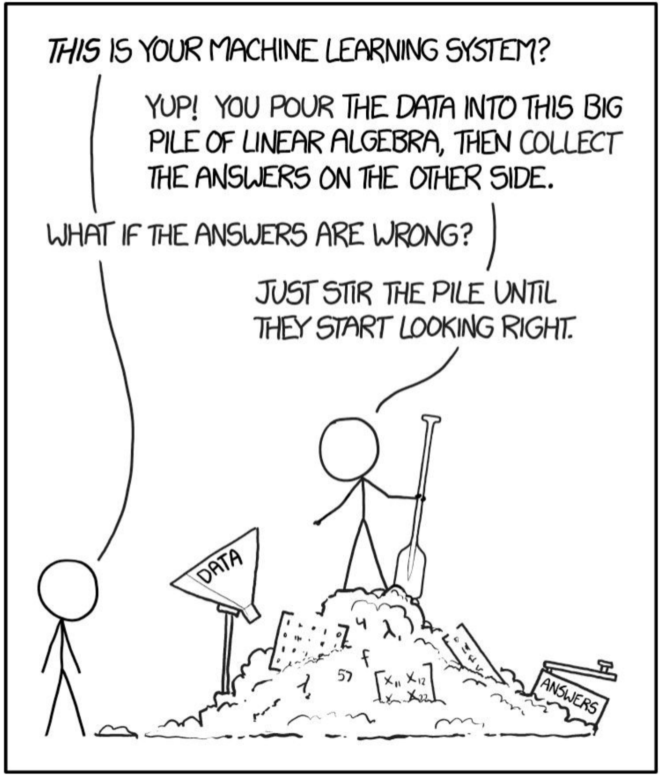
predicted label



cat

... learn to **predict the label** associated to a new example

Supervised Learning



Supervised learning: classification

examples **labels**

cat

dog

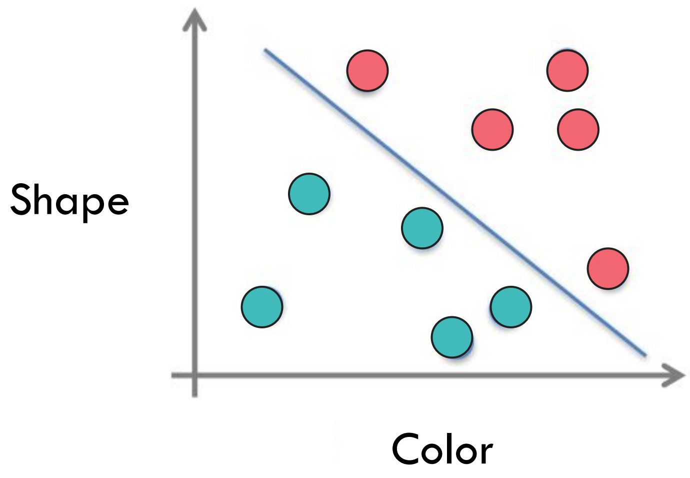
cat

Classification: a finite set of labels

Classification

- Given a **training set** $T = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots (\mathbf{x}_N, y_N)\}$
- Learn a **function** to predict y given $\mathbf{x}$
- $\mathbf{x}$ is generally **multidimensional** (multiple features)

$$f : \mathbb{R}^d \to \{1, 2, \ldots, k\}$$



Classification Applications

- **Face recognition**
- **Character recognition**
- **Spam detection**
- **Medical diagnosis:** from symptoms to illnesses
- **Biometrics:** recognition/authentication using physical or behavioral characteristics (face, iris …)

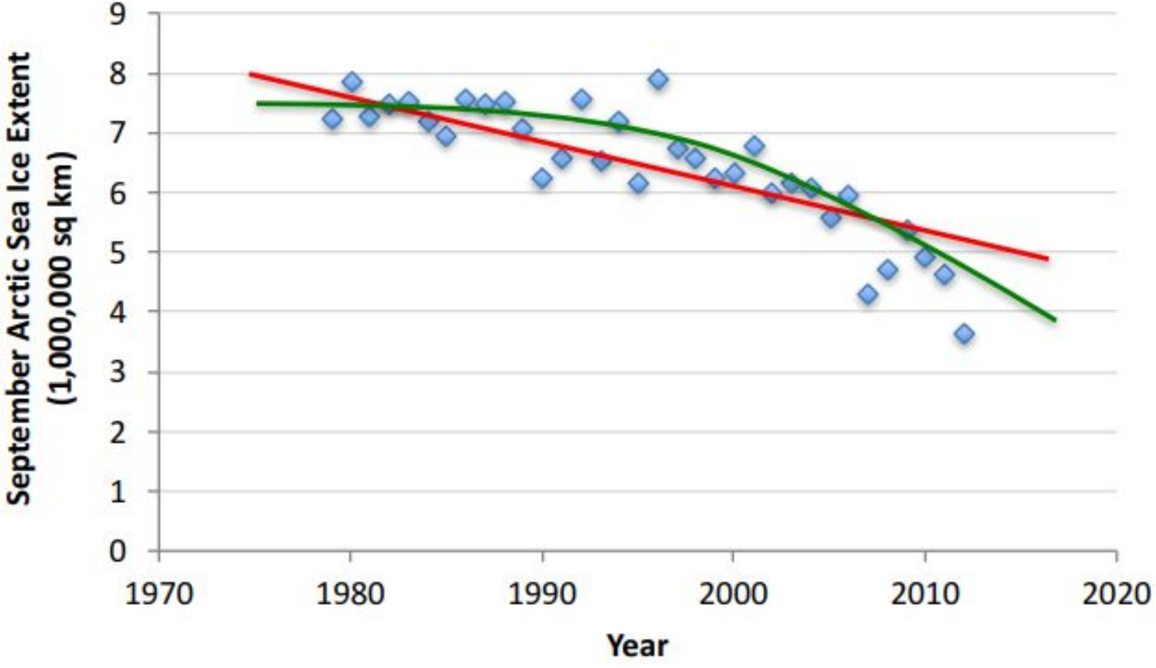
…

Supervised learning: regression

examples	labels
	*5kg*
	*10kg*
	*5kg*

Regression: Label is real-valued

Regression Example

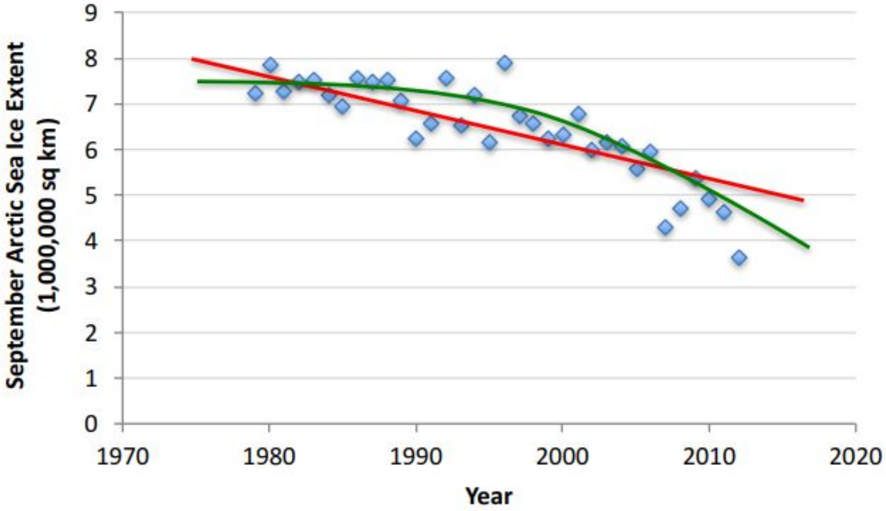


$\mathbf{x}$: year $\quad\quad\quad$ y : ice-extent

Regression

- Given a **training set** $T=\{(\mathbf{x}_1,y_1), (\mathbf{x}_2\ ,y_2), \ldots\ (\mathbf{x}_N,y_N)\}$
- Learn a **function** to predict y given $\mathbf{x}$
- y is real valued

$$f : \mathbb{R}^d \rightarrow \mathbb{R}$$



Regression Applications

- **Economics and finance:** predict the value of a stock
- **Car navigation:** angle of the steering wheel, acceleration, …
- **Temporal trends:** weather over time

…

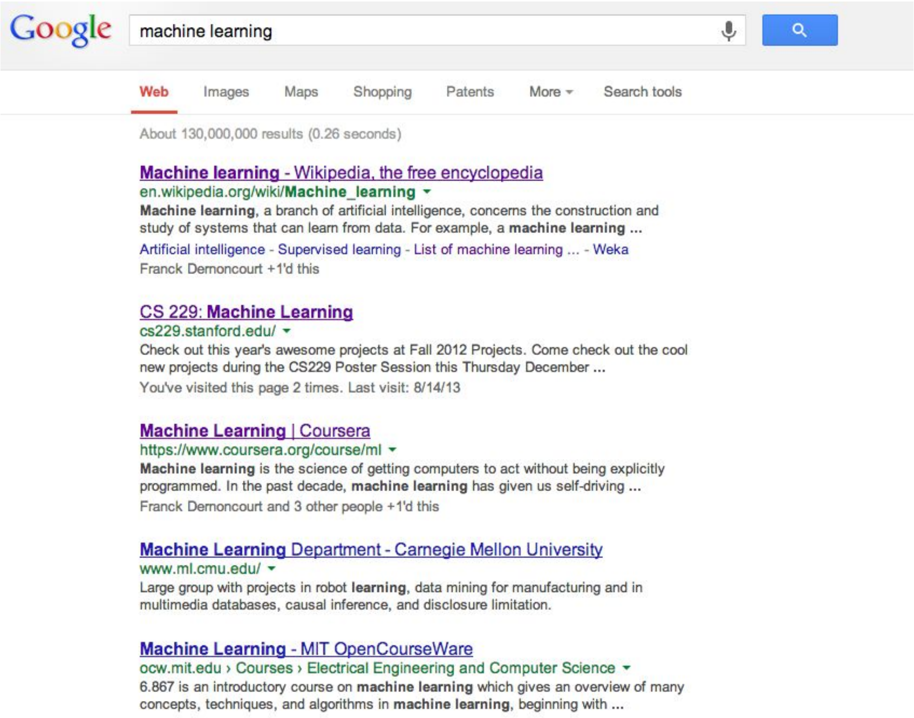
Supervised learning: ranking

examples	labels
	2
	1
	3

Ranking: label indicates an order

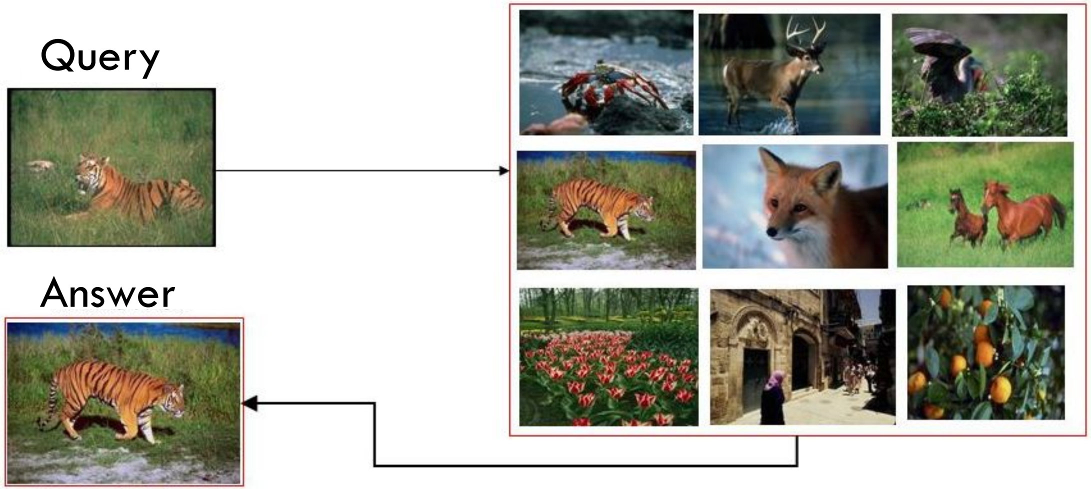
Ranking example

Given a query and a set of web pages, rank them according to relevance



Ranking example

Given a query image, find the most visually similar images (with an order) in the database.



Ranking Applications

- **User preference,** e.g. Netflix movie ranking
- Image **retrieval**
- Flight **search** (search in general)

…

Supervised Learning Summary

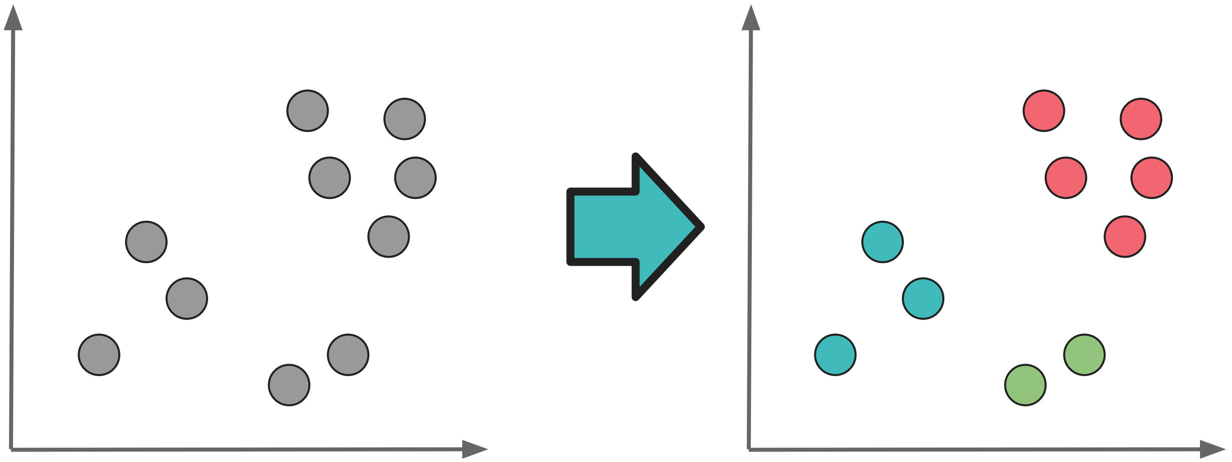
- **Classification**:
  - Categorize input data into predefined classes or categories.
- **Regression**:
  - Predict continuous numerical values based on input features.
- **Ranking**:
  - Order a set of items based on their relevance or preference.

Unsupervised learning

Unsupervised learning: given data, i.e. examples, but no labels

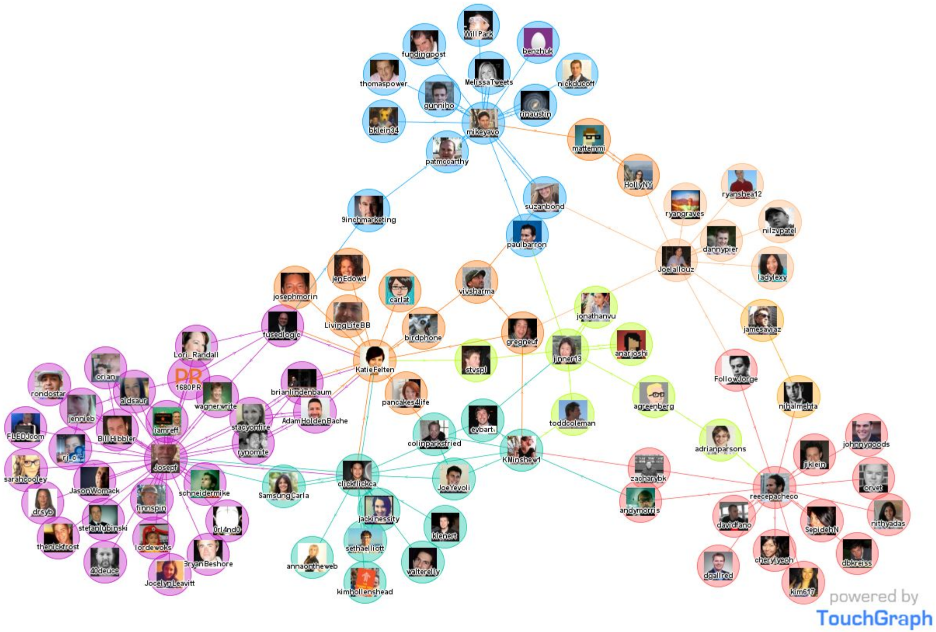
Clustering

Given $T = \{\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_N\}$ (without labels) output hidden structure behind the x's, that is the clusters



Clustering applications

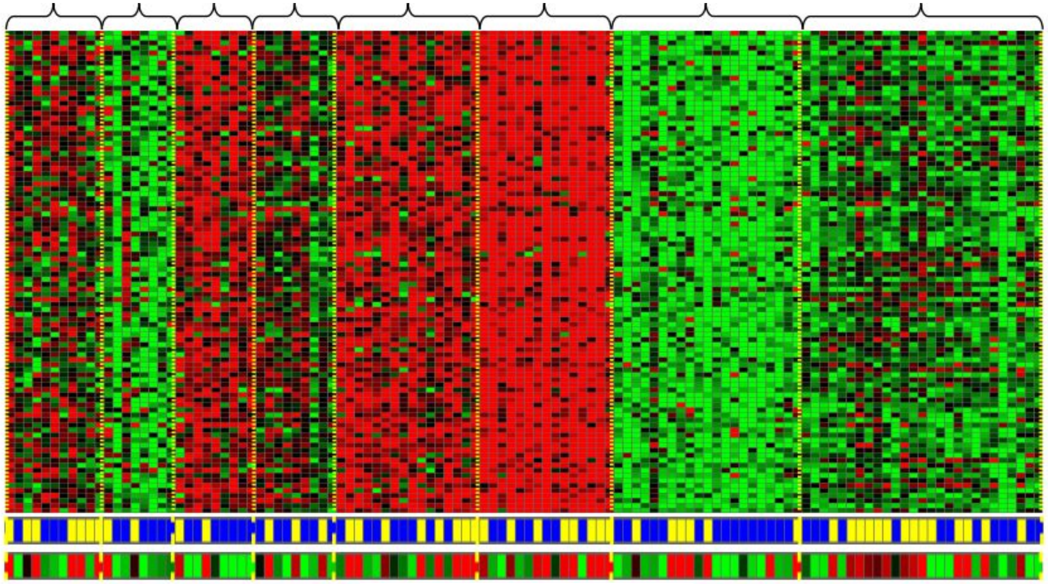
Social network analysis



Clustering applications

Genomics: group individuals by genetic similarity

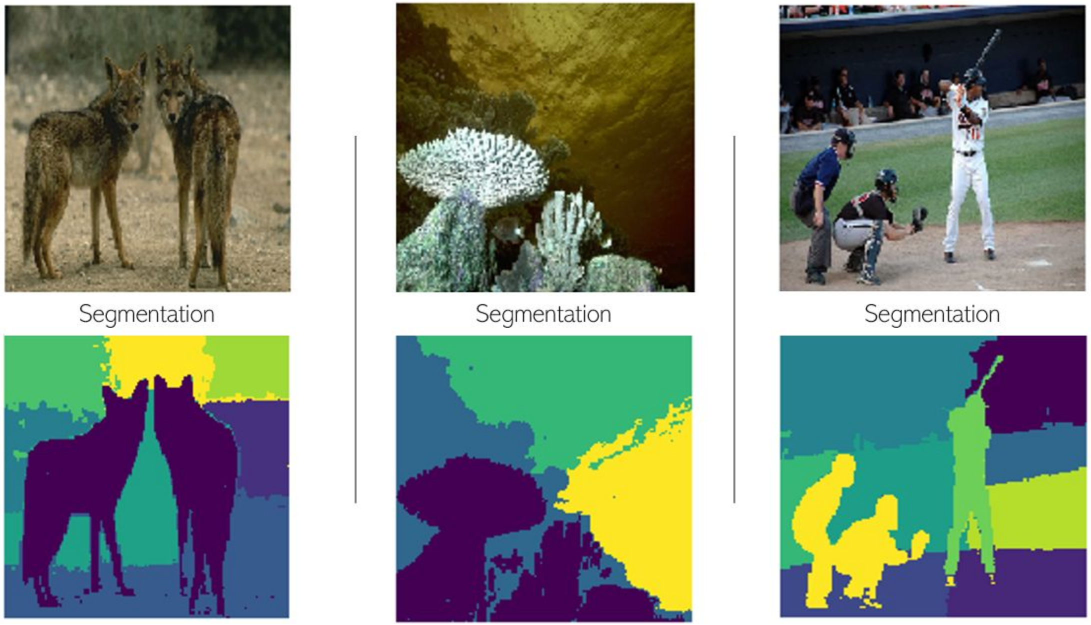
Genes



Individuals

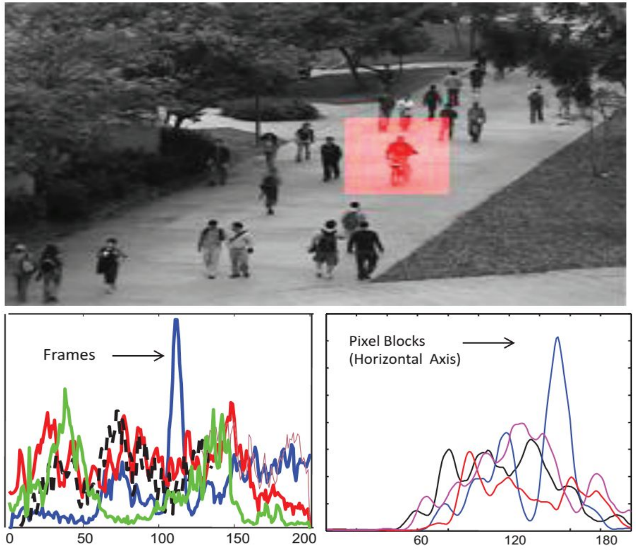
Clustering applications

Image segmentation



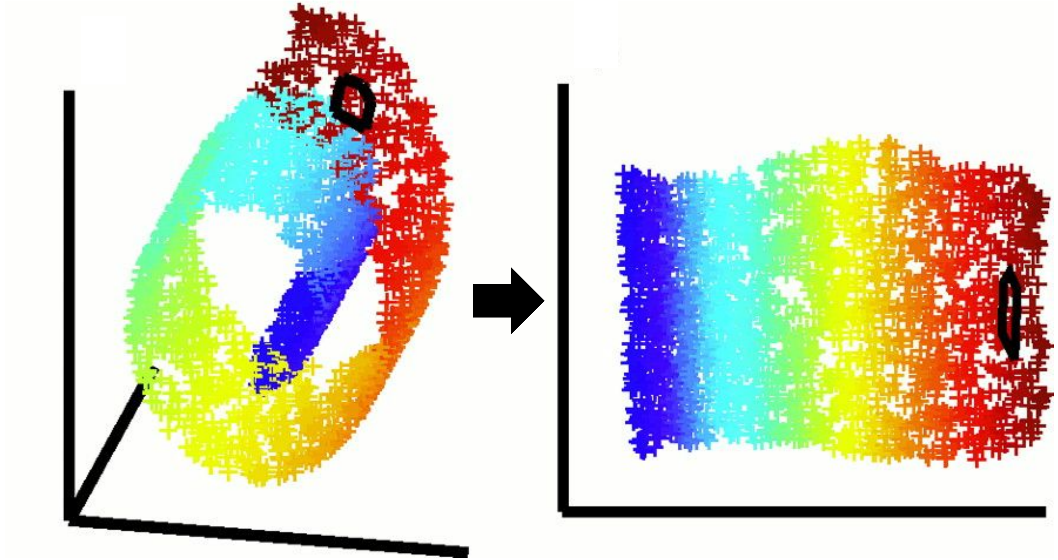
Anomaly detection

- Analyze a set of events or objects and flags some of them as being unusual or atypical.
- Example: video surveillance



Dimensionality Reduction

Reduce the number of features under consideration by mapping data into another low dimensional space.

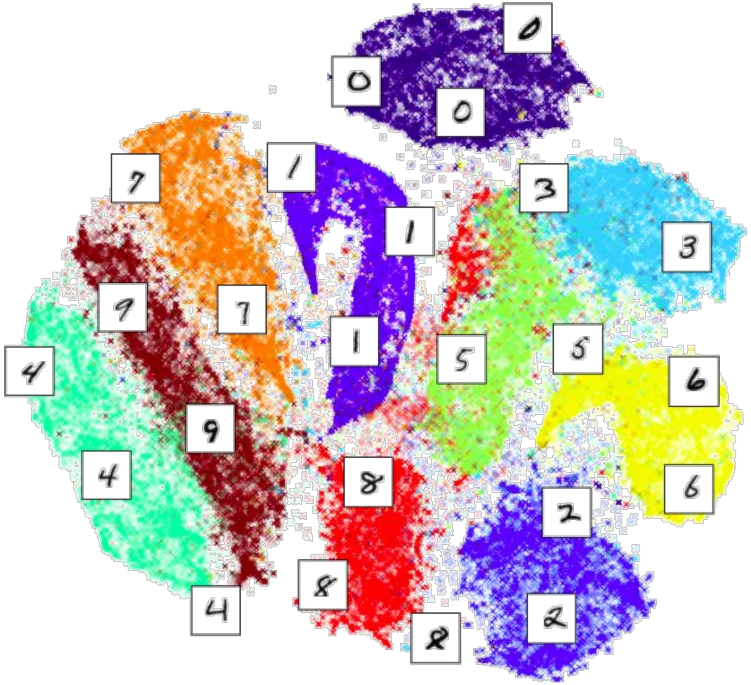


$$f : \mathbb{R}^d \rightarrow \mathbb{R}^m$$

$$m << d$$

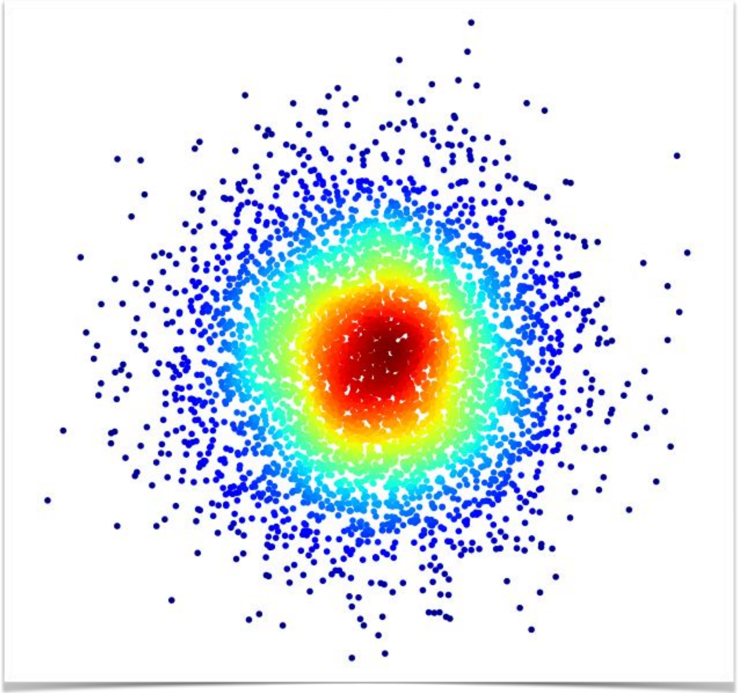
Dimensionality Reduction applications

Inspect your classification algorithm



Density estimation

Find a probability distribution that fits your training data $T = \{\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_N\}$

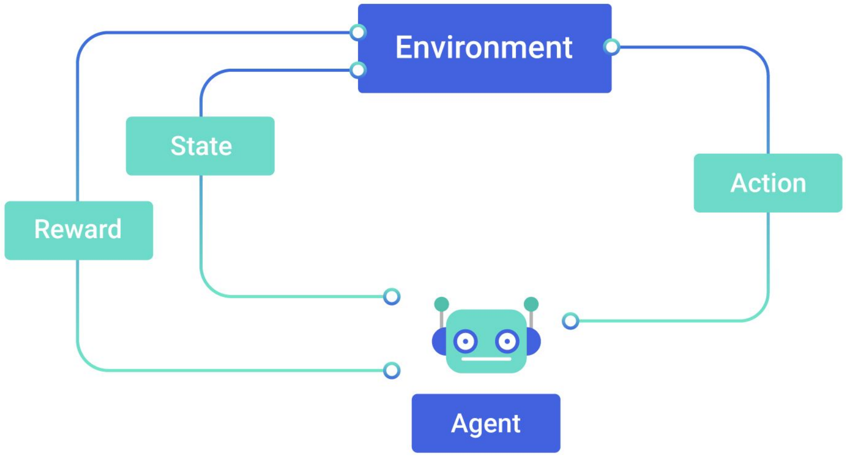


Unsupervised Learning Summary

- **Clustering**:
  - Groups similar data points into clusters or groups based on their inherent characteristics or features.
- **Dimensionality Reduction**:
  - Reduces the number of features in a dataset while preserving the most important information
- **Density estimation**:
  - Find a probability distribution that fits the data

Reinforcement Learning

Idea: an **agent** learns from the **environment** by interacting with it and receiving **rewards** for performing **actions**.



Reinforcement Learning example

Agent: A small robot on a grid

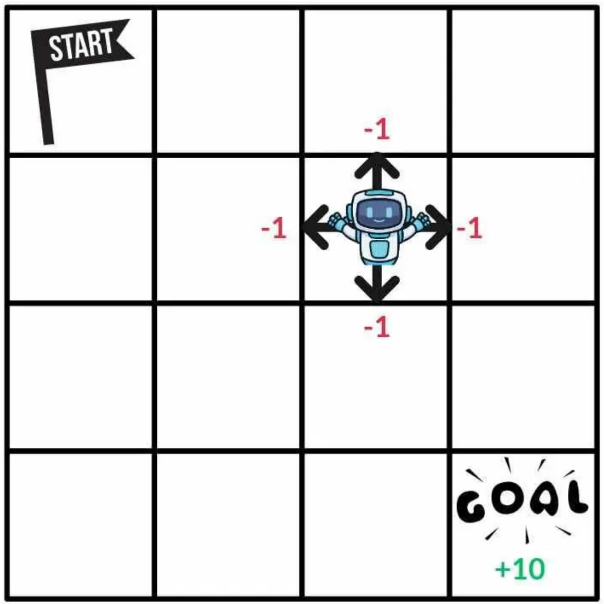
Environment: The grid world with a goal cell

State: Robot position on the grid

Actions: Move up, down, left, right

Reward: +10 when it reaches the goal, -1 otherwise

Learning idea: The robot tries many paths, gets rewarded when it finds the goal and gradually learns the shortest route.



Types of Learning: Summary

Supervised Learning

- Learn from labeled data
- Examples: spam detection, image classification

Unsupervised Learning

- Learn from unlabeled data to discover patterns or structure
- Examples: clustering, dimensionality reduction

Reinforcement Learning

- Learn by trial and error with rewards, choose actions that maximize long-term reward
- Examples: game playing, robot navigation

Types of Learning: Summary

Other learning variations

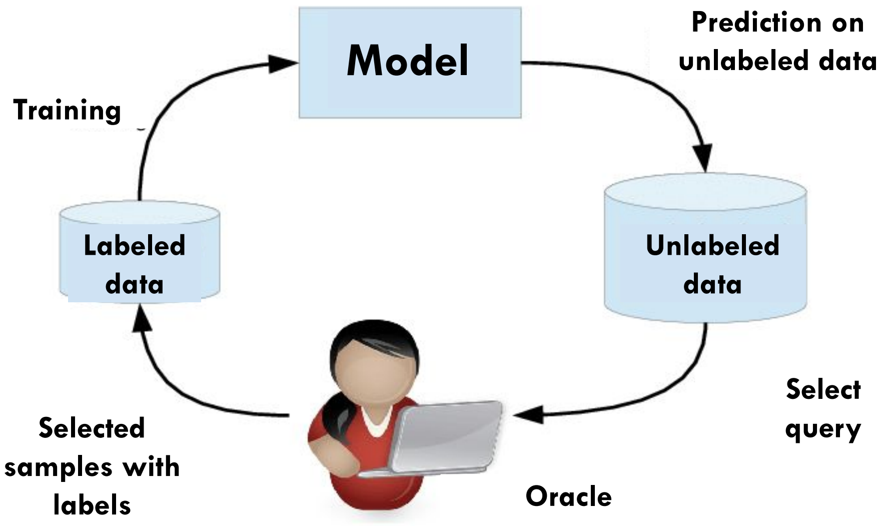
We saw supervised, unsupervised and reinforcement learning

Other settings: **semi-supervised**

Other learning variations

We saw supervised, unsupervised and reinforcement learning

Other settings: **active learning**



Online vs. Offline (batch) learning

- How are we getting the data: **online vs. offline (batch) learning**

- The difference between online and batch (offline) learning lies in **how the learning process occurs** and **how the model is updated with new data.**

Online vs. Offline (batch) learning

Batch (Offline) Learning:

- The model learns from the entire dataset in one go and is updated only after processing all the data.
- Once the model is trained, it does not change.
- Typically used when the dataset fits into memory, can be processed efficiently as a whole.

Online Learning:

- The model learns incrementally from each new data point or small batches of data.
- Allows the model to adapt to changes in the data distribution over time.
- Suitable for scenarios where data arrives sequentially and needs to be processed in real-time or where computational resources are limited.

Back to Data and Features

examples **features**

grey, small, ...

brown, big, ...

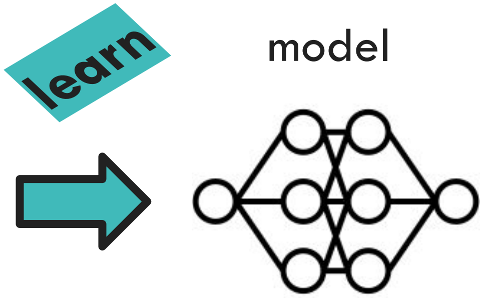
black, small, ...

Back to Data and Features

- How our algorithms actually “view” the data
- **Features are** the **questions we can ask about the examples**
- Features in generally are **represented with vectors**

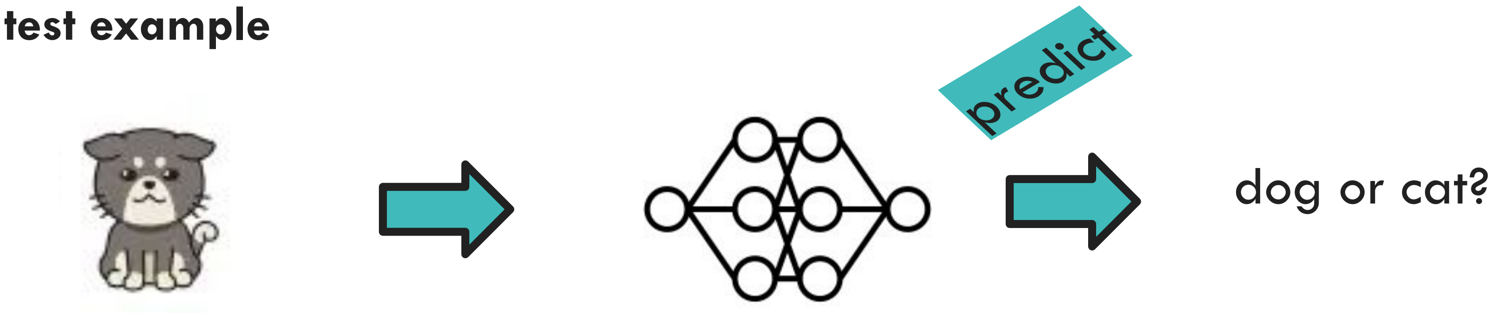
Classification revisited

features	**labels**
grey, small, …	*cat*
brown, big, …	*dog*
black, small, …	*cat*



During training, **learn a model** of what distinguishes dogs and cats **based on the features**

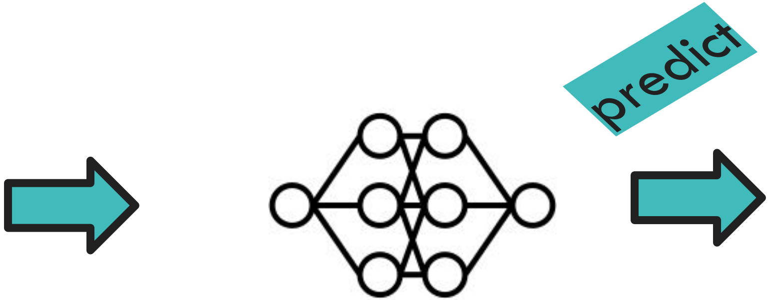
Classification revisited



Classification revisited

test example

grey, small …



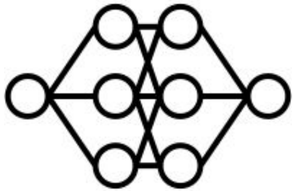
dog or cat?

The new example is described **by the features**

Classification revisited

test example

grey, small …



predict

cat

The model can then classify a new example based on **the features**

Classification revisited

Training data

examples	label
grey, small, ...	*cat*
brown, big, ...	*dog*
black, small, ...	*cat*

Test data

grey, small ... ?

Classification revisited

Training data

examples	label
grey, small, …	*cat*
brown, big, …	*dog*
black, small, …	*cat*

Test data

grey, small … ?

Classification revisited

Training data

examples	label
grey, small, …	*cat*
brown, big, …	*dog*
black, small, …	*cat*

Test data

grey, small … ?

Learning is about **generalizing** from the training data

Generalization

- **Generalization** in machine learning refers to the ability of a trained model to perform well on new, unseen data that was not used during the training process.

- A model generalizes well when it can accurately make predictions or produce outputs for data it has not encountered before.

Generalization

How do we achieve generalization?

What do we assume about the training and test set?

Training & Test Set

Training data

Test set

Training and test set needs to be “similar”

Training & Test Set

Training data

Test set

Not always the case, then learning is not possible!

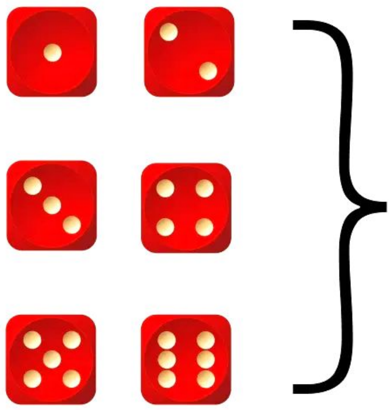
More technically...

- We are going to use the **probabilistic model** of learning
- There is some probability distribution over example & label pairs called the **data generating distribution**
- **Both the training data and the test set** are generated based on this distribution

Recap: Probability distribution

Describes **how likely (i.e. probable) certain events are**

- Considers probabilities for all possible events
- Probabilities are between 0 and 1 (inclusive)
- Sum of probabilities over all events is 1



$$\frac{1}{6}$$

Recap: Probability distribution

- A **probability distribution** is a mathematical function that describes the likelihood of different outcomes or events occurring within a certain set of possible outcomes.
- It provides a way to represent and **quantify uncertainty** in various situations.
- In simpler terms, a probability distribution tells us **how probable it is for different events** to occur.
- It assigns probabilities to each possible outcome, with the sum of all probabilities equaling 1.

Recap: Probability distribution

- Types of Probability Distributions
  - **Discrete Probability Distribution** (for discrete random variables): the probability is assigned to specific values.

    Example: The Bernoulli distribution (outcomes: 0 or 1)

  - **Continuous Probability Distribution** (for continuous random variables): the probability is described over an interval of values.

    Example: The Normal distribution.

Discrete Probability distribution

For a **discrete random variable** X, probabilities are given by the **Probability Mass Function (PMF)**

$$P(X = x_i) = p(x_i) \qquad \sum p(x_i) = 1$$

specific values

Continuous Probability distribution

For a **continuous random variable** X, probabilities are given by the **Probability Density Function (PDF)** $f(x)$

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

Key property

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

Normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Probability distribution

Training data

High probability	Low probability
grey cat	*brown cat*
brown dog	*white dog*
…	…

DATA GENERATING DISTRIBUTION

Training data

Test set

data generating distribution

DATA GENERATING DISTRIBUTION

- A **data generating distribution** refers to the underlying probability distribution that generates the observed data points in a dataset.
- The data generating distribution captures the inherent patterns, structure, and noise present in the data.
- The **data generating distribution is unknown.** We only have access to his indirect representation of the training set.

Summary

- The learning process
- Data & features
- Supervised, unsupervised and reinforcement learning
- Training and test data, generalization, data generating distribution

QUESTIONS?

Additional Resources

- A visual introduction to machine learning

  https://narrative-flow.github.io/exploratory-study-1